

COURSE Project

Introduction of AI in Healthcare

Professional Master's in AI in Healthcare | Term 2

Intelligent Hospital Information System (I-HIS) Using Vibe Coding and AI

Project Objective

Students will work in groups of 10 to build an Intelligent Hospital Information System (I-HIS) using Python, Flask, and AI-assisted coding tools such as ChatGPT, Claude, Gemini, Cursor, and GitHub Copilot.

The project focuses on understanding how AI can support healthcare professionals through clinical decision support, diagnostic reasoning, medical imaging, patient education, and multi-agent collaboration.

Each week, students will add a new AI medical specialist to the hospital system.

Week 1

Introduction to AI in Healthcare

Students will begin building a basic Healthcare Information System (HIS) using the Flask framework.

Required Tasks

1. Create a Flask project and configure the development environment.
2. Design and implement a Patient Registration Form.
3. Create an Electronic Patient Record (EPR) module to store patient information.
4. Build the system Homepage and Navigation Menus.
5. Create a personal GitHub account (if not already available).
6. Create a dedicated GitHub repository for the Healthcare Information System project.
7. Upload and maintain the project source code in the GitHub repository.
8. Deploy and run the HIS application from the GitHub repository.
9. Share the GitHub repository link with the instructor for review and assessment.

AI Specialist Added

Hospital Receptionist AI

The first AI module integrated into the HIS will simulate an intelligent hospital receptionist.

Functions

- Register new patients.

- Collect patient demographic information.
- Record chief complaints and reasons for visit.
- Direct patients to the appropriate healthcare service or department.
- Provide basic guidance and appointment information.

Deliverables

By the end of Week 1, each student must submit:

- A functioning Flask-based Healthcare Information System (HIS) prototype.
- A working Patient Registration module.
- An Electronic Patient Record module.
- A GitHub repository containing the complete project source code.
- Evidence that the HIS application runs successfully from the repository.
- The GitHub repository link shared with the instructor.

Expected Outcome

A running Intelligent Healthcare Information System (HIS) prototype integrated with a basic Hospital Receptionist AI module and managed through GitHub version control.

Week 2

Fundamentals of Machine Learning in Healthcare and Dentistry

Tasks

- Add disease prediction module.
- Use a simple healthcare dataset.
- Display prediction results.

AI Specialist Added

General Practitioner AI

Functions

- Predict common diseases.
- Assess patient risk factors.
- Recommend routine investigations.

Deliverable

General medical consultation module.

Week 3

Clinical Decision Support Systems (CDSS)

Tasks

- Develop clinical decision support dashboard.

- Create alert system for abnormal findings.

AI Specialist Added

ICU Specialist AI

Functions

- Monitor vital signs.
- Detect critical conditions.
- Recommend urgent interventions.

Deliverable

ICU monitoring and alert system.

Week 4

AI in Medical Imaging

Tasks

- Add medical image upload feature.
- Analyze X-ray images using AI tools.

AI Specialist Added

Radiologist AI

Functions

- Interpret chest X-rays.
- Detect pneumonia.
- Detect possible fractures.

Deliverable

Medical imaging module.

Week 5

Generative AI in Healthcare

Tasks

- Build healthcare chatbot.
- Implement Retrieval-Augmented Generation (RAG).

AI Specialist Added

Patient Education AI

Functions

- Explain diseases.
- Explain medications.
- Answer patient questions.

Deliverable

Interactive healthcare assistant.

Week 6

AI in Drug Discovery and Pharmacology

Tasks

- Create medication recommendation module.
- Add drug interaction checker.

AI Specialist Added

Clinical Pharmacist AI

Functions

- Check drug interactions.
- Suggest medications.
- Flag contraindications.

Deliverable

Medication safety system.

Week 7

AI in Mental Health

Tasks

- Develop mental health assessment form.
- Analyze patient responses.

AI Specialist Added

Psychiatrist AI

Functions

- Depression screening.
- Anxiety screening.
- Mental health recommendations.

Deliverable

Mental health support module.

Week 9

Introduction to AI Search and Clinical Reasoning

Tasks

- Build symptom-to-diagnosis reasoning engine.
- Create differential diagnosis workflow.

AI Specialist Added

Clinical Reasoning AI

Functions

- Generate differential diagnoses.
- Rank likely diseases.
- Explain diagnostic logic.

Deliverable

Clinical reasoning module.

Week10

BFS in Emergency Medicine

Tasks

- Implement Breadth First Search diagnostic tree.

AI Specialist Added

Emergency Physician AI

Functions

- Emergency triage.
- Explore multiple diagnostic possibilities.
- Prioritize urgent conditions.

Deliverable

Emergency decision support system.

Week 11

DFS in Oncology and Complex Diagnosis

Tasks

- Implement Depth First Search diagnostic pathway.

AI Specialist Added

Oncologist AI

Functions

- Analyze cancer-related symptoms.
- Explore complex diagnostic pathways.
- Suggest investigations and staging.

Reinforcement Learning in Healthcare

Tasks

- Simulate treatment optimization.

- Compare alternative treatment plans.

AI Specialist Added

Treatment Planning AI

Functions

- Recommend treatment strategies.
- Optimize patient management pathways.
- Evaluate treatment outcomes.

Deliverable

Treatment optimization module.

Oncology reasoning module.

Week 12

Agentic and Multi-Agent AI in Hospital Ecosystems

Tasks

- Connect all AI specialists together.
- Enable communication between agents.
- Generate integrated clinical reports.

AI Specialists Collaborating

- Receptionist AI
- GP AI
- ICU AI
- Radiologist AI
- Pharmacist AI
- Psychiatrist AI
- Clinical Reasoning AI
- Emergency AI
- Oncologist AI
- Treatment Planning AI

Deliverable

Complete Intelligent Hospital Information System.

Week 13

Final Presentation and Demonstration

Live Demonstration

Students present a complete patient case.

The system should:

1. Register the patient.

2. Record symptoms.
3. Analyze uploaded medical data.
4. Generate differential diagnoses.
5. Recommend investigations.
6. Suggest treatment options.
7. Produce a final clinical report.

Presentation Requirements

- Project overview.
- AI tools used.
- Prompts used during development.
- Medical datasets used.
- Clinical validation results.
- Strengths and limitations.
- Ethical considerations.
- Future improvements.

Final Submission

- Source code.
- Project report.
- Presentation slides.
- Demonstration video.
- User manual.

Expected Learning Outcomes

By the end of the project, students will be able to:

1. Explain major AI applications in healthcare.
2. Use Vibe Coding tools to create healthcare applications.
3. Evaluate AI-generated medical recommendations.
4. Understand clinical decision support systems.
5. Apply search algorithms to diagnostic reasoning.
6. Understand reinforcement learning concepts in healthcare.
7. Design multi-agent healthcare AI systems.
8. Critically assess AI safety, ethics, and limitations in medicine.